SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005 & ANSI/NCSL Z540-1-1994

ADVANCED TECHNOLOGY SERVICES – CATERPILLAR TECHNICAL CENTER
14009 Old Galena Rd
Mossville, IL 61552
John Goodrich Phone: 309 494 5123

CALIBRATION

Valid To: August 31, 2017 Certificate Number: 1592.05

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory at the location above as well as the one satellite laboratory location listed below to perform the following calibrations[1]:

I. Chemical

Parameter/Equipment	Range	CMC[2] (±)	Comments
Gas Detection Equipment			
Calibration of Total Nitrous Oxide (NO/NOx Analyzers)	(0 to 500) parts in 10^6 (0 to 2500) parts in 10^6 (0 to 10 000) parts in 10^6	2.6 % volume 2.6 % volume 2.6 % volume	Standard gases and Horiba GDC-703 gas divider
Calibration of Total Hydrocarbon/Methane Analyzers	(0 to 500) parts in 10^6 (0 to 2500) parts in 10^6 (0 to 10 000) parts in 10^6	2.6 % volume 2.6 % volume 2.6 % volume	
Calibration of Total Hydrocarbon/Propane Analyzers	(0 to 500) parts in 10^6 (0 to 2500) parts in 10^6 (0 to 10 000) parts in 10^6	2.6 % volume 2.6 % volume 2.6 % volume	
Calibration of Oxygen Analyzers	(0 to 25 000) parts in 10^6	2.6 % volume	
Calibration of Carbon Monoxide Analyzers	(0 to 5000) parts in 10^6 (0 to 16 000) parts in 10^6	2.6 % Volume 2.6 % Volume	
Calibration of Carbon Dioxide Analyzers	(0 to 160 000) parts in 10^6	2.6 % Volume	

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Gas Detection Equipment (cont) –			Standard gases and Horiba GDC-703 gas divider
Particulate Exhaust	(2 to 200) SLPM	0.29 % + 0.059 SLPM	Sierra BG-3 and LFE

II. Dimensional Testing/Calibration[5]

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Surface Analysis[5]	Ra up to 200 µin	23 µin (0.58 µm)	Mitutoyo SVC-3000
	Ra up to 200 µin	22 µin (0.56 µm)	Mitutoyo SVC-638
	Ra up to 5.0 µm	0.40 µm	Zeiss Surfcom SD2000
Differential Transducers[5] (LVDT)	Up to 2 in	0.017 %	Mitutoyo 197-201 w/ Agilent 34410A
Form[5] –			
Roundness	(3 to 350) mm	1.4 µm	Talyrond 3
Straightness	Up to 700 mm	3.4 µm	
Flatness	Up to 350 mm	10 µm	
Inside Diameter[5]	(55 to 75) mm	4.3 µm	Dial bore #4 gage
	(75 to 150) mm	5.3 µm	Dial bore #5 gage
	(150 to 210) mm	6.6 µm	Dial bore #6 gage

Parameter/Equipment	Range	CMC[2, 6] (±)	Comments
Length[5] – 1D	Up to 25.4 mm	1.1 µm	Mitutoyo digital indicator
	Up to 600 mm	29 µm	Tesa height gage
	Up to 750 mm	2.2 mm	Chatillon test stand
Length[5] – 3D	*X* axis: 200 mm *Y* axis: 250 mm *Z* axis: 75 mm	64 µm	Dynascope microscope
	X axis: 200 mm *Y* axis: 250 mm *Z* axis: 75 mm	64 µm	Dynascope multi lens
External Diameter[5]	Up to 25.4 mm	6.4 µm	Mitutoyo micrometer
	(25.4 to 50.8) mm	8.1 µm	Starrett 733 micrometer
	(50.8 to 76.2) mm	7.0 µm	
	(76.2 to 101.6) mm	6.6 µm	
	(101.6 to 127) mm	8.1µm	
	(127 to 152.4) mm	8.1 µm	
	(0.1 to 101.6) mm	(4.2 + 0.00089D) µm	Laser micrometer

III. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
DC Voltage – Generate	(0 to 220) mV (0.22 to 2.2) V (2.2 to 11) V (11 to 22) V (22 to 220) V (220 to 1100) V (500 to 5000) V	9.3 µV/V + 0.60 µV 7.2 µV/V + 1.0 µV 7.1 µV/V + 3.5 µV 7.4 µV/V + 6.5 µV 8.2 µV/V + 80 µV 9.2 µV/V + 0.50 mV 0.17 % + 2.5 V	Fluke 5700A SRS-PS350
DC Voltage – Measure	(0 to 200) mV (0.2 to 2) V (2 to 20) V (20 to 200) V (200 to 1000) V	5.3 µV/V + 0.10 µV 3.7 µV/V + 0.40 µV 3.8 µV/V + 4.0 µV 5.6 µV/V + 40 µV 5.7 µV/V + 0.50 mV	Fluke 8508A
DC Current – Generate	(0 to 220 µA (0.22 to 2.2) mA (2.2 to 22) mA (22 to 220) mA (0.22 to 2.2) A (2.2 to 11) A (11 to 20.5) A (0 to 2) A (2 to 20) A (20 to 120) A 120 A to 2.5 kA	54 µA/A + 8.0 nA 52 µA/A + 8.0 nA 52 µA/A + 80 nA 62 µA/A + 0.80 µA 95 µA/A + 25 µA 3.6 mA/A + 0.48 mA 0.17 % + 0.75 mA 0.0094 % + 0.080 mA 0.011 % + 0.80 mA 0.009 % + 4.8 mA 0.74 % + 0.84 A	Fluke 5700A Fluke 5725A Fluke 5522A- SC1100 Fluke 52120A w/ 25/50 turn Coil
DC Current – Measure	(10 to 200) µA (0.2 to 2) mA (2 to 20) mA (20 to 200) mA (0.2 to 2) A (2 to 20) A	14 µA/A + 0.40 nA 16 µA/A + 4.0 nA 18 µA/A + 40 nA 57 µA/A + 0.80 µA 0.021 % + 16 µA 0.047 % + 0.40 mA	Fluke 8508A

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
High DC Current – Generate	(100 to 1000) A (1000 to 3000) A	0.42 A 0.82 A	Current amplifier w/ transducers
Capacitance – Generate	(0.19 to 0.39999) nF (0.4 to 1.0999) nF (1.1 to 3.2999) nF (3.3 to 10.9999) nF (11 to 32.9999) nF (33 to 109.999) nF (110 to 329.999) nf (0.33 to 1.09999) µF (1.1 to 3.2999) µF (3.3 to 10.9999) µF (11 to 32.9999) µF	2.8 % + 10 pF 1.3 % + 10 pF 0.60 % + 10 pF 0.27 % + 10 pF 0.26 % + 0.10 nF 0.27 % + 0.10 nF 0.27 % + 0.30 nF 0.27 % + 1.0 nF 0.27 % + 3.0 nF 0.27 % + 10 nF 0.41 % + 30 nF	Fluke 5522A-SC1100

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Voltage – Generate			
(2.2 to 22) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.059 % + 5.0 µV 0.027 % + 5.0 µV 0.023 % + 5.0 µV 0.046 % + 5.0 µV 0.095 % + 7.0 µV 0.14 % + 12 µV 0.20 % + 25 µV 0.44 % + 25 µV	Fluke 5700A
(22 to 220) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.057 % + 13 µV 0.024 % + 8.0 µV 0.013 % + 8.0 µV 0.034 % + 8.0 µV 0.087 % + 25 µV 0.11 % + 25 µV 0.17 % + 35 µV 0.36 % + 80 µV	

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Voltage – Generate (cont)			
(0.22 to 2.2) V	(10 to 20) Hz	0.051 % + 80 µV	Fluke 5700A
	(20 to 40) Hz	0.017 % + 25 µV	
	40 Hz to 20 kHz	0.0097 % + 6.0 µV	
	(20 to 50) kHz	0.013 % + 16 µV	
	(50 to 100) kHz	0.026 % + 70 µV	
	(100 to 300) kHz	0.047 % + 0.13 mV	
	(300 to 500) kHz	0.11 % + 0.35 mV	
	500 kHz to 1 MHz	0.24 % + 0.85 mV	
(2.2 to 22) V	(10 to 20) Hz	0.068 % + 0.80 mV	
	(20 to 40) Hz	0.018 % + 0.25 mV	
	40 Hz to 20 kHz	0.0097 % + 60 µV	
	(20 to 50) kHz	0.013 % + 0.16 mV	
	(50 to 100) kHz	0.026 % + 0.35 mV	
	(100 to 300) kHz	0.054 % + 1.5 mV	
	(300 to 500) kHz	0.13 % + 4.3 mV	
	500 kHz to 1 MHz	0.29 % + 8.5 mV	
(22 to 220) V	(10 to 20) Hz	0.053 % + 8.0 mV	
	(20 to 40) Hz	0.019 % + 2.5 mV	
	40 Hz to 20 kHz	0.0089 % + 0.80 mV	
	(20 to 50) kHz	0.023 % + 3.5 mV	
	(50 to 100) kHz	0.051 % + 8.0 mV	
(220 to 1100) V	50 Hz to 1 kHz	84 µV/V + 3.5 mV	Fluke 5700A w/ 5725A
	40 Hz to 1 kHz	0.095 % + 4.0 mV	
	(1 to 20) kHz	0.17 % + 6.0 mV	
	(20 to 30) kHz	0.60 % + 11 mV	
(220 to 750) V	(30 to 50) kHz	0.42 % + 11 mV	
	(50 to 100) kHz	1.6 % + 45 mV	

Parameter / Equipment	Range	CMC[2, 4] (±)	Comments
AC Voltage – Measure			
(0 to 200) mV	(10 to 40) Hz (40 to 100) Hz 100 Hz to 2 kHz (2 to 10) kHz (10 to 30) kHz (30 to 100) kHz	0.016 % + 4.0 µV 0.013 % + 4.0 µV 0.012 % + 2.0 µV 0.015 % + 4.0 µV 0.035 % + 8.0 µV 0.077 % + 20 µV	Fluke 8508A
(0.2 to 2) V	(10 to 40) Hz (40 to 100) Hz 100 Hz to 2 kHz (2 to 10) kHz (10 to 30) kHz (30 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz	0.013 % + 20 µV 0.0094 % + 20 µV 0.0079 % + 20 µV 0.011 % + 20 µV 0.023 % + 40 µV 0.058 % + 0.20 mV 0.30 % + 2.0 mV 1.0 % + 20 mV	
(2 to 20) V	(10 to 40) Hz (40 to 100) Hz 100 Hz to 2 kHz (2 to 10) kHz (10 to 30) kHz (30 to 100) kHz (100 to 300) kHz 300 kHz to 1 MHz	0.014 % + 0.20 mV 0.0097 % + 0.20 mV 0.0084 % + 0.20 mV 0.012 % + 0.20 mV 0.023 % + 0.40 mV 0.058 % + 2.0 mV 0.30 % + 20 mV 1.0 % + 0.20 V	
(20 to 200) V	(10 to 40) Hz (40 to 100) Hz 100 Hz to 2 kHz (2 to 10) kHz (10 to 30) kHz (30 to 100) kHz	0.014 % + 2.0 mV 0.0096 % + 2.0 mV 0.0082 % + 2.0 mV 0.012 % + 2.0 mV 0.023 % + 4.0 mV 0.058 % + 20 mV	
(200 to 1000) V	40 Hz to 10 kHz (10 to 30) kHz	0.012 % + 21 mV 0.026 % + 42 mV	

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Current – Generate			
(9 to 220) uA	(10 to 20) Hz (20 to 40) Hz (40 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	0.072 % + 25 nA 0.037 % + 20 nA 0.018 % + 16 nA 0.069 % + 40 nA 0.17 % + 80 nA	Fluke 5700A
(0.22 to 2.2) mA	(10 to 20) Hz (20 to 40) Hz (40 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	0.072 % + 40 nA 0.037 % + 35 nA 0.017 % + 35 nA 0.062 % + 0.40 µA 0.16 % + 0.80 µA	
(2.2 to 22) mA	(10 to 20) Hz (20 to 40) Hz (40 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	0.075 % + 0.40 µA 0.037 % + 0.35 µA 0.017 % + 0.35 µA 0.062 % + 4.0 µA 0.16 % + 8.0 µA	
(22 to 220) mA	(10 to 20) Hz (20 to 40) Hz (40 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	0.075 % + 4.0 µA 0.036 % + 3.5 µA 0.018 % + 3.5 µA 0.062 % + 40 µA 0.16 % + 80 µA	
(0.22 to 2.2) A	(20 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	0.067 % + 35 µA 0.079 % + 80 µA 0.87 % + 0.16 mA	
(0 to 2) A	(10 to 65) Hz (65 to 300) Hz 300 Hz to 1 kHz (1 to 3) kHz (3 to 6) kHz (6 to 10) kHz	0.013 % + 1.1 mA 0.023 % + 1.1 mA 0.078 % + 1.1 mA 0.23 % + 9.3 mA 0.78 % + 25 mA 1.6 % + 62 mA	Fluke 52120A w/ 25/50 turn coil
(2 to 20) A	(10 to 65) Hz (65 to 300) Hz 300 Hz to 1 kHz (1 to 3) kHz (3 to 6) kHz (6 to 10) kHz	0.013 % + 9.4 mA 0.024 % + 9.4 mA 0.078 % + 9.4 mA 0.23 % + 31 mA 0.78 % + 62 mA 2.3 % + 94 mA	

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Current – Generate (cont)			
(20 to 120) A	(10 to 65) Hz (65 to 300) Hz 300 Hz to 1 kHz (1 to 3) kHz (3 to 6) kHz (6 to 10) kHz	0.014 % + 19 mA 0.024 % + 28 mA 0.079 % + 94 mA 0.23 % + 0.23 A 0.78 % + 0.42 A 3.1 % + 0.70 A	Fluke 52120A w/ 25 turn coil
120 A to 3 kA	(10 to 65) Hz (65 to 300) Hz	0.72 % + 0.84 A 0.72 % + 0.84 A	
120 A to 1 kA (120 to 300) A (0 to 100) A (0 to 50) A	300 Hz to 1 kHz (1 to 3) kHz (3 to 6) kHz (6 to 10) kHz	0.71 % + 0.84 A 0.80 % + 1.2 A 1.5 % + 1.2 A 5.0 % + 1.2A	
AC Current – Measure			
(0 to 200) µA	10 Hz to 10 kHz	0.052 % + 20 nA	Fluke 8508A
(0.1 to 2.0) mA	10 Hz to 10 kHz	0.030 % + 0.20 µA	
(2.0 to 20) mA	10 Hz to 10 kHz	0.031 % + 2.0 µA	
(20 to 200) mA	100 Hz to 5 kHz	0.030 % + 20 µA	
(0.2 to 2.0) A	100 Hz to 2 kHz (6 to 10) kHz	0.063 % + 0.20 mA 0.074 % + 0.20 mA	
(0.2 to 2.0) A	100 Hz to 2 kHz (2 to 10) kHz	0.83 % + 2.0 mA 2.5 % + 2.0 mA	

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Oscilloscopes –			
DC Signal			
50 Ω Load	(-6.6 to 6.6) V	0.33 % + 40 µV	Fluke 5522A-SC1100
1 MΩ Load	(-130 to 130) V	0.12 % + 40 µV	
Amplitude Square Wave			
50 Ω ,10 Hz to 10 kHz	± 1 mV to 6.6 Vp-p	0.25 % + 40 µV	
1 MΩ			
10 Hz to 1 kHz	± 1 mV to 130 Vp-p	0.10 + 40 µV	
(1 to 10) kHz	± 1 mV to 130 Vp-p	0.25 + 40 µV	
Leveled Sine Wave (Into 50 Ω Load)	50 kHz Reference	2.0 % + 300 µV	
Flatness @ 50 kHz Reference	50 kHz to 100 MHz	1.7 % + 100 µV	
	(100 to 300) MHz	2.2 % + 100 µV	
	(300 to 600) MHz	4.1 % + 100 µV	
	(600 to 1100) MHz	5.3 % + 100 µV	
Time Markers (50 Ω Load)	5 s to 50 ms	(58 + 1000t) µs/s	t = time in seconds
	20 ms to 2 ns	5.7 µs/s	
Resistance – Generate, Fixed Points	0 Ω	55 µΩ	Fluke 5700A
	1 Ω	0.10 mΩ	
	1.9 Ω	0.18 mΩ	
	10 Ω	0.30 mΩ	
	19 Ω	0.58 mΩ	
	100 Ω	1.0 mΩ	
	190 Ω	3.3 mΩ	
	1 kΩ	13 mΩ	
	1.9 kΩ	25 mΩ	
	10 kΩ	0.12 Ω	
	19 kΩ	0.23 Ω	
	100 kΩ	1.4 Ω	
	190 kΩ	2.7 Ω	
	1 MΩ	20 Ω	
	1.9 M	41 Ω	
	10 MΩ	0.42 kΩ	
	19 M Ω	0.94 kΩ	
	100 MΩ	12 kΩ	

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Resistance – Measure	(0 to 10) Ω (10 to 100) Ω (0.1 to 1.0) kΩ (1.0 to 10) kΩ (10 to 100) kΩ (0.1 to 1.0) MΩ (1 to 10) MΩ (10 to 100) MΩ	21 µΩ/Ω + 50 µΩ 18 µΩ/Ω + 0.50 mΩ 16 µΩ/Ω + 0.50 mΩ 16 µΩ/Ω + 5.0 mΩ 16 µΩ/Ω + 50 mΩ 22 µΩ/Ω + 2.0 Ω 63 µΩ/Ω + 0.10 kΩ 0.058 % + 1.0 kΩ	HP 3458A, option II
	(0 to 2) Ω (2 to 20) Ω (20 to 200) Ω (0.2 to 2.0) kΩ (2.0 to 20) kΩ (20 to 200) kΩ (0.2 to 2.0) MΩ (2 to 20) MΩ (20 to 200) MΩ (0.2 to 2) GΩ	61 µΩ/Ω + 4.0 µΩ 12 µΩ/Ω + 14 µΩ 9.4 µΩ/Ω + 50 µΩ 9.4 µΩ/Ω + 0.5 mΩ 9.4 µΩ/Ω + 5.0 mΩ 9.5 µΩ/Ω + 50 mΩ 11 µΩ/Ω + 1.0 Ω 24 µΩ/Ω + 0.10 kΩ 1.4 % + 10 kΩ 0.18 % + 0.10 MΩ	Fluke 8508A
Electrical Calibration of Thermocouple Indicators & Indicating Systems			
Type E	(-250 to -100) °C (-100 to -25) °C (-25 to 350) °C (350 to 650) °C (650 to 1000) °C	0.50 °C 0.16 °C 0.14 °C 0.16 °C 0.21 °C	Fluke 5522A-SC1100
Type J	(-210 to -100) °C (-100 to -30) °C (-30 to 150) °C (150 to 760) °C (760 to 1200) °C	0.27 °C 0.16 °C 0.14 °C 0.17 °C 0.23 °C	
Type K	(-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1000) °C (1000 to 1372) °C	0.33 °C 0.18 °C 0.16 °C 0.26 °C 0.40 °C	

Parameter/Equipment	Range	CMC[2,] (±)	Comments
Electrical Calibration of Thermocouple Indicators & Indicating Systems (cont)			
Type T	(-250 to -150) °C (-150 to 0) °C (0 to 120) °C (120 to 400) °C	0.63 °C 0.24 °C 0.16 °C 0.14 °C	Fluke 5522A-SC1100

IV. Fluid Quantities

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Fuel Flow – Measure	(25 to 15000) g/min	0.41 % + 0.59 g/min	Gravimetric calibrator g = grams
Air Flow – Measure	(20 to 23000) kg/hr (1.3 to 5500) kg/hr	0.49 % 0.33 %	Subsonic nozzles , critical flow nozzles
Liquid Flow – Measuring Equipment	(0 to 1514) l/min	2.4 %	FTI Omni Track Gravimetic (OT-400)
Water Flow – Measuring Equipment	(71 to 728.5) kg/min	0.11 % + 0.10 kg/min	CMF200 Micro Motion w/ Agilent 34970A

V. Mechanical

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Scales	(1 to 121) kg	0.0041 % + 1.3 g	Standard weights

Parameter/Equipment	Range	CMC[2, 3] (±)	Comments
Force –			
Tension	(50 to 150) N	2.7 N	C-K Engineering (tangential tension)
	(50 to 500) N	2.4 N	Chatillon load cell (diametric tension)
Compression & Tension	(0 to 4450) N (0 to 11 120) N (0 to 22 241) N (0 to 44 482) N (0 to 111 205) N	6.7 N 17 N 34 N 67 N 170 N	Comparison with standard load cell
Compression	(0 to 667 233) N (0 to 2 224 110) N	1.8 kN 5.5 kN	Comparison with standard load cell
Pressure –			
Hydraulic	500 kPa to 5 MPa (5 to 500) MPa	0.069 % + 15 Pa 0.096 % + 64.9 Pa	DH Instruments PG7302 AMH
Pneumatic	(0 to 62.2) kPa (>62.2 to 206.9) kPa (104 to 2064.7) kPa (>2064.7 to 6890.3) kPa	0.059 % + 34 Pa 0.049 % + 21 Pa 0.055 % + 27 Pa 0.056 % + 1.1 Pa	DH Instruments PPC4
Rotational Speed	(50 to 5000) RPM	1.4 RPM	Phototach and 6680B frequency counter
Torque – Measuring Equipment	(1 to 8605.9) Nm	0.017 % + 0.044 N·m	Lever arm w/ Class F weights
Torque Wrenches	(4.5 to 20) Nm (27 to 122) Nm (162 to 732) Nm	0.14 Nm 2.2 Nm 3.3 Nm	Precision torque transducer

Parameter/Equipment	Range	CMC[2] (±)	Comments
Torque – Shaft	(1695 to 3389) Nm (3390 to 8474) Nm (8475 to 16 948) Nm (16 949 to 33 895) Nm (33 896 to 84 739) Nm	17 Nm 36 Nm 73 Nm 0.015 kNm 0.026 kNm	Torque shaft cal bench with arm

VI. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature, RTD – Measuring Equipment	(-40 to 99) °C (100 to 150) °C	0.069 °C 0.046 °C	SPRT and temperature bath
Dew Point – Measuring Equipment	(0 to 25) °C	0.20 °C	Thunder Scientific 1200

VII. Time & Frequency

Parameter/Equipment	Range	CMC[2] (±)	Comments
Frequency – Measuring Equipment	1 kHz to 80 MHz	27 parts in 10^6	Agilent 33250A
Frequency – Measure	(0.01 to 225) MHz	7.0 x 10^{-6} Hz	Fluke PM6680B

[1] This laboratory offers commercial dimensional testing/calibration service.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of k = 2. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] In the statement of CMC, a percent refers to a percent of reading unless otherwise noted.

[4] The measurands stated are generated using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure the measurand in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a fraction/percentage of the reading plus a fixed floor specification.

[5] This laboratory meets *R205 – Specific Requirements: Calibration Laboratory Accreditation Program* for the types of dimensional tests listed above and is considered equivalent to that of a calibration.

[6] In the statement of CMC, *D* is the numerical value of the diameter. In the statement of CMC, *L* is the numerical value of the nominal length.

[7] This accreditation covers testing/calibration performed at the main laboratory listed above, and the following satellite laboratory listed below.

ADVANCED TECHNOLOGY SERVICES – CATERPILLAR TECHNICAL CENTER
AC 6132 Dock 29 Rench Rd Route 29
Mossville, IL 61552
John Goodrich   Phone: 309 494 5123

CALIBRATION

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory at the location below to perform the following calibrations[1]:

I. Electrical – DC/Low Frequency

Parameter	Range	CMC[2] (±)	Comments
Voltage – Measure	(0.25 to 9.5) V	0.49 V	FAST UNIT model 3525
Frequency – Measure	(0.01 to 225) MHz	0.15 mHz	FAST UNIT model 3525

II. Thermodynamics

Parameter	Range	CMC[2] (±)	Comments
Temperature – Measure	(-40 to 120) °C	0.92 °C	Class B RTD with Fluke 1586A
Humidity – Measuring Equipment	(10 to 90) % RH	4 % RH	Omega IBTHX-SD

[1] This laboratory offers commercial calibration services.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of k = 2. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

Accredited Laboratory

A2LA has accredited

ADVANCED TECHNOLOGY SERVICES – CATERPILLAR TECHNICAL CENTER

Mossville, IL

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSLI Z540-1-1994 and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

AMERICAN ASSOCIATION FOR LABORATORY ACCREDITATION
CORPORATE SEAL 1978
DISTRICT OF COLUMBIA
A2LA

Presented this 20th day of October 2015.

President & CEO
For the Accreditation Council
Certificate Number 1592.05
Valid to August 31, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.